

# **Note on *A course in Arithmetic* by J.P. Serre**

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# 1 Chapter 1

This is a list of proofs of nontrivial results in Chapter 1 of *A course in Arithmetic* by J.P. Serre.

Let  $K$  be a finite field with  $q$  elements.

## Theorem 1.1

Let  $u$  be an integer  $\geq 0$ . The sum

$$S(X^u) = \sum_{x \in K} x^u = \begin{cases} -1 & \text{if } u > 0 \text{ and } (q-1) \mid u \\ 0 & \text{otherwise} \end{cases}$$

$u = 0$  is subtle here. We define  $x^0 = 1$  for all  $x \in K$  (this definition is nice when we work with polynomials because we can write 1 as  $x^0$  to maintain symmetry), so  $S(X^0) = q = 0$ .

The proof is easy. Use primitive roots.

The variant proof:  $x \mapsto x^u$  is a group homomorphism from  $K^\times$  to itself. The kernel is all  $x$  such that  $x$  is a root of  $x^{\gcd(u, q-1)} - 1$ . So the image has size  $\frac{q-1}{\gcd(u, q-1)}$ , which implies the image is just the  $d = \frac{q-1}{\gcd(u, q-1)}$  roots of unity. So the sum is 0 unless  $d = 1$  (sum of  $d$ -th roots of unity is 0 if  $d \geq 2$  is coprime to  $p$ ).

## Theorem 1.2 (Chevalley-Warning Theorem)

Let  $P_1, \dots, P_r$  be polynomials in  $n$  variables over a finite field  $K$  of characteristic  $p$ . If the sum of their degrees is less than  $n$ , then the number of common solutions to the equations  $P_1 = \dots = P_r = 0$  is divisible by  $p$ .

*Proof.* This uses a common trick in counting roots over finite fields. Consider the polynomial

$$Q = \prod_{i=1}^r (1 - P_i^{q-1})$$

The number of common solutions is equal to  $\sum_{x \in K^n} Q(x)$  since if  $P_i(x) = 0$  for all  $i$ , then  $Q(x) = 1$ , otherwise  $Q(x) = 0$ .

Expand this. For each monomial in the expansion  $x_1^{u_1} \dots x_n^{u_n}$ , since  $\sum \deg(P_i)(q-1) < n(q-1)$ , we have  $u_j$  is not divisible less than  $q-1$  for some  $j$ . So by the previous theorem, the sum over  $K^n$  of this monomial is 0. (that's why we define  $x^0 = 1$  above). So the number of common solutions is 0 mod  $p$ .  $\square$

### Corollary 1.3

All quadratic forms in at least 3 variables over a finite field have nontrivial zeros.

### Theorem 1.4

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$$

*Proof.* Let  $\alpha$  be the primitive 8th root of unity  $\alpha$  in  $\overline{\mathbb{F}_p}$ . Then,  $y^2 = (\alpha + \alpha^{-1})^2 = 2$ . If  $p \equiv \pm 1 \pmod{8}$ , then  $y^p = y$ , so 2 is a square in  $\mathbb{F}_p$ . If  $p \equiv \pm 3 \pmod{8}$ , then  $y^p = -y$ , so 2 is not a square in  $\mathbb{F}_p$ .  $\square$

### Theorem 1.5 (Law of Quadratic Reciprocity)

For two odd primes  $p$  and  $q$ , we have

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

The key is to consider the Gauss sum: Let  $\zeta$  be a primitive  $p$ -th root of unity in  $\overline{\mathbb{F}_p}$ , then define

$$y = \sum_{x \in \mathbb{F}_p} \left(\frac{x}{p}\right) \zeta^x$$

Then we have  $y^2 = \left(\frac{-1}{p}\right) p = (-1)^{\frac{p-1}{2}} p$  and  $y^{q-1} = \left(\frac{q}{p}\right)$ . Combining these two gives the result.

## 2 Meeting 1 on Jan. 28 2026

This is a list of nontrivial results in the first meeting of the course.

### Theorem 2.1

The finite subgroup of a field's multiplicative group is cyclic.

This can be easily proven using the structure theorem of finite abelian groups and the fundamental theorem of Algebra.

**Definition 2.2.** For any prime  $p$  or  $p = \infty$ , we define the

$$\|x\|_p = \begin{cases} p^{-v_p(x)} & p \text{ is a prime} \\ \|x\| & p = \infty \end{cases}$$

### Theorem 2.3

For any rational number  $r$ ,  $\prod_p \|r\|_p = 1$

This is trivial from the definition of  $v_p$ .

We can in general define a norm on a number field  $K$  as follows:

**Definition 2.4.**  $\|\cdot\| : K \rightarrow \mathbb{R}_{\geq 0}$  is a norm if

1.  $\|x\| = 0$  iff  $x = 0$
2.  $\|xy\| = \|x\| \|y\|$
3.  $\|x + y\| \leq \|x\| + \|y\|$

### Corollary 2.5

$\|1\| = 1$  and  $\|-1\| = 1$

**Definition 2.6.** Norm  $N_1$  and  $N_2$  are equivalent if there exists a real number  $c$  such that  $N_2(x) = N_1(x)^c$  for all  $x$ .

### Theorem 2.7 (Ostrowski's Theorem)

Any non-trivial norm on  $\mathbb{Q}$  is equivalent to either the usual absolute value  $\|\cdot\|_\infty$  or to  $\|\cdot\|_p$  for some prime  $p$ .

*Proof.*

□

### Theorem 2.8 (Hensel's Lemma)

*Proof.*

□

### Theorem 2.9 (Hasse principle)

### Example 2.10

$f(x, y, z) = 5x^2 + 7y^2 - 13z^2$ . We want to find a nontrivial  $f(x, y, z) = 0$  has non-trivial solutions in  $\mathbb{Q}$  using Hensel's Lemma.

Consider this curve in  $\mathbb{P}^3$ , is there any other solution?

Use parameterization of this curve.

### Theorem 2.11

If  $n \geq 3$ ,  $x^n + y^n = z^n$  has no non-trivial solutions in  $\mathbb{C}[x]$ .

## 3 Chapter 2

### 3.1 $\mathbb{Z}_p$ and $\mathbb{Q}_p$

Let  $A_n = \mathbb{Z}/p^n\mathbb{Z}$ . There is a natural projection map  $\phi_n : A_n \rightarrow A_{n-1}$ .

**Definition 3.1.** We define  $\mathbb{Z}_p$  as the limit (in category of rings) of

$$\cdots \xrightarrow{\phi_{n+1}} A_n \xrightarrow{\phi_n} A_{n-1} \xrightarrow{\phi_{n-1}} \cdots \xrightarrow{\phi_2} A_1$$

By the definition, any  $x \in \mathbb{Z}_p$  can be represented as a sequence  $(\dots, x_2, x_1)$  where  $x_n \in A_n$  and  $\phi_n(x_n) = x_{n-1}$ . Alternatively, we can represent  $x$  as an infinite sum

$$x = a_0 + a_1p + a_2p^2 + \cdots$$

where  $a_i \in \{0, 1, \dots, p-1\}$ .

We can define the topology on  $\mathbb{Z}_p$  as the induced topology from the product topology on  $\prod A_n$  where  $A_n$  is given the discrete topology.

### Theorem 3.2

$\mathbb{Z}_p$  is compact.

*Proof.* We will need a lemma from topology.

### Lemma 3.3

**Closed subset of a compact space is compact.**

*Proof.* If  $K$  is compact and  $F \subseteq K$  is closed, then for any open covering of  $F$ , we can add  $K \setminus F$  to get an open covering of  $K$ . So there is a finite subcovering, removing  $K \setminus F$  gives a finite subcovering of  $F$ .  $\square$

Since  $\prod A_n$  is compact by Tychonoff's theorem, it suffices to show  $\mathbb{Z}_p$  is closed in  $\prod A_n$ . It suffices show that its complement is open. For any  $y \notin \mathbb{Z}_p$ , there exists some  $j$  such that  $\phi_j(y_j) \neq y_{j-1}$ . There exists an open set  $U = \prod U_n$

where  $U_n = A_n$  for  $n \neq j, j-1$ ,  $U_j = \{y_j\}$  and  $U_{j-1} = \{y_{j-1}\}$ . It's clear that  $U$  is an open neighborhood of  $y$  and  $U \cap \mathbb{Z}_p = \emptyset$ . So  $\mathbb{Z}_p$  is closed.  $\square$

### Theorem 3.4

$$0 \rightarrow A_m \xrightarrow{\times p^n} A_{m+n} \xrightarrow{\phi_{m+n,n}} A_n \rightarrow 0$$

is an exact sequence of abelian groups.

This is easy to check. The first injective map is moving the the last  $m$  digits  $\times p^n$  to the left by  $n$  places and fill the last  $n$  digits with zero, the second map is just truncating the last  $n$  digits. We will use this in  $\mathbb{Z}_p$ .

### Theorem 3.5

$$0 \rightarrow \mathbb{Z}_p \xrightarrow{\times p^n} \mathbb{Z}_p \rightarrow A_n \rightarrow 0$$

is an exact sequence of abelian groups.

*Proof.* First, we need to show  $\mathbb{Z}_p \xrightarrow{\times p^n} \mathbb{Z}_p$  is injective. Notice that if  $\phi_{m+n,n}(x_{m+n}) = x_n$ , then  $p^m x_n$  and  $p^m x_{m+n}$  are the same in  $A_{m+n}$ . Therefore, we conclude that multiplying by  $p^n$  is just adding  $n$  zeros to the end, that is sending  $(\dots, x_2, x_1)$  to  $(\dots, p^n x_2, p^n x_1, 0, \dots, 0)$ . Let  $j$  be the smallest integer such that the  $j$ -th digit of  $x$  is nonzero (not divisible by  $p^j$ ), then the  $j+n$ -th digit of  $\times p^n x$  is also nonzero since it's not divisible by  $p^{j+n}$ , so the map is injective.

The image of  $\times p^n$  is just  $p^n \mathbb{Z}_p$ , which is exactly the kernel of the projection map to  $A_n$  since the last  $n$  digits are all zero.  $\square$

### Theorem 3.6

Two propositions:

1. For an element  $x \in \mathbb{Z}_p$  to be invertible, it is equivalent to say that it's not divisible by  $p$ , i.e., the first digit  $a_0$  in its expansion is nonzero.
2. Let  $U$  be the group of units in  $\mathbb{Z}_p$ . Then we have the decomposition. For any  $x \in \mathbb{Z}_p$ , we can write

$$x = p^n u$$

where  $n \geq 0$  and  $u \in U$  in a unique way.

For part 1, if  $x_1$  is invertible,  $a_i$  are coprime to  $p$ . By Bezout's lemma, there exists  $a_i, b_i \in \mathbb{Z}$  such that  $a_i x_i + b_i p^i = 1$ . Let  $a = (\dots, a_2, a_1)$ , this is clearly

the inverse of  $x$  and we can check it's indeed in  $\mathbb{Z}_p$ .

Other parts are easy.

**Definition 3.7.** We define  $v_p(x) = n$  if  $x = p^n u$  where  $u$  is a unit in  $\mathbb{Z}_p$ . We define  $v_p(0) = +\infty$ .

### Proposition 3.8

$v_p(xy) = v_p(x) + v_p(y)$  and  $v_p(x + y) \geq \inf\{v_p(x), v_p(y)\}$  with equality if  $v_p(x) \neq v_p(y)$ .

This allows us to define a metric on  $\mathbb{Z}_p$  as follows:

**Definition 3.9.**  $d(x, y) = p^{-v(x-y)}$

A notable property is that  $d(x, y) = d(x - z, y - z)$  for any  $x, y, z \in \mathbb{Z}_p$ . So the metric is translation invariant. In particular, we have  $d(x, y) = d(0, y - x)$ .

### Theorem 3.10

The metric space  $(\mathbb{Z}_p, d)$  is homeomorphic to the topological space  $\mathbb{Z}_p$  defined as the subspace of  $\prod A_n$  with the product topology.

*Proof.* It suffices to show that the identity map from  $(\mathbb{Z}_p, d)$  to  $\mathbb{Z}_p$  is continuous in both directions.

For any open set  $U \subseteq \mathbb{Z}_p$ ,

$$U = \mathbb{Z}_p \cap \left( \bigcup_{i \in I} \left( \prod U_{n,i} \right) \right) = \bigcup_{i \in I} \left( \left( \prod U_{n,i} \cap \mathbb{Z}_p \right) \right)$$

where  $U_{n,i} = A_n$  for all but finitely many  $n$ .

What does  $T_i = \prod U_{n,i} \cap \mathbb{Z}_p$  look like? Let  $N$  be the largest integer such that  $U_{N,i} \neq A_N$ . Then,  $T_i$  is the set of  $x$  such that the first  $N$  digits of  $x$  are fixed and the rest can be anything possible, that is  $x = (\dots, a_N, \dots, a_1)$  where  $a_1, \dots, a_N$  are fixed. The preimage of  $T_i$  in  $(\mathbb{Z}_p, d)$  is precisely the open ball centered at  $x$  with radius  $p^{-N}$ , that is  $x + p^N \mathbb{Z}_p$ .

Conversely, what does any open ball in  $(\mathbb{Z}_p, d)$  look like in  $\mathbb{Z}_p$ ? First, the open ball of 0 is just ideals  $p^n \mathbb{Z}_p$ . For other point, if the radius of the open ball is  $p^{-N}$ , for some  $N \in \mathbb{Z}$ .  $d(x, y) = d(0, x - y) < p^{-N}$  means  $v_p(x - y) > N$ , which means the first  $N$  digits of  $x$  and  $y$  are the same. So the open balls are the cosets of  $p^N \mathbb{Z}_p$ . So the preimage of this open ball is the set of  $y$  such that the first  $N$  digits are fixed, which is clearly open in  $\mathbb{Z}_p$ .  $\square$



**Remark 3.11.** Any open ball is also closed in  $\mathbb{Z}_p$ , since the image of  $d(0, x)$  is discrete in  $\mathbb{R}$ . So  $\mathbb{Z}_p$  is totally disconnected.

**Remark 3.12.**  $\mathbb{Z}_p$  is complete under this metric. This follows from the fact that  $\mathbb{Z}_p$  is compact.

**Remark 3.13.**  $\mathbb{Z}$  is dense in  $\mathbb{Z}_p$ . Just find an integer with the same first  $n$  digits as the element in  $\mathbb{Z}_p$  you want to approximate.

**Definition 3.14.** We define  $\mathbb{Q}_p$  as the field of fractions of  $\mathbb{Z}_p$ . Equivalently, it's the localization of  $\mathbb{Z}_p$  at  $U = 1, p, p^2, \dots$

**Definition 3.15.** For any  $x \in \mathbb{Q}_p$ , we can write  $x = p^n u$  where  $n \in \mathbb{Z}$  and  $u \in U$ . We define  $v_p(x) = n$  and  $v_p(0) = +\infty$ .

$x \in \mathbb{Q}_p$  is in  $\mathbb{Z}_p$  iff  $v_p(x) \geq 0$ .

### Proposition 3.16

propositions about  $\mathbb{Q}_p$ :

1. The metric  $d(x, y) = p^{-v_p(x-y)}$  makes  $\mathbb{Q}_p$  a topological field.
2.  $\mathbb{Z}_p$  is a clopen subset of  $\mathbb{Q}_p$ .
3.  $\mathbb{Q}_p$  is locally compact, that is for any  $x \in \mathbb{Q}_p$ , there exists open set  $U$  and compact set  $K$  such that  $x \in U \subseteq K$ .
4.  $\mathbb{Q}$  is dense in  $\mathbb{Q}_p$ .

For any open neighborhood of  $x \in \mathbb{Q}_p$ , it looks like  $x + p^n \mathbb{Z}_p$  for some  $n$  (in fact clopen). Notice that translation and scaling are homeomorphisms, so any neighborhood of  $x$  is compact. Others are easy to check.

### Proposition 3.17

The metric  $d$  is an ultrametric, that is it satisfies the strong triangle inequality

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}$$

for any  $x, y, z \in \mathbb{Q}_p$ .

As a result,  $d(u_m, u_n) \leq \sup_{k \geq n} d(u_k, u_{k+1})$  for any sequence  $\{u_n\}$  in  $\mathbb{Q}_p$  and any  $m > n$ . Therefore, a sequence  $\{u_n\}$  is Cauchy iff  $d(u_n, u_{n+1}) \rightarrow 0$  as  $n \rightarrow \infty$  iff  $\lim_{n \rightarrow \infty} u_{n+1} - u_n = 0$ .